

Meeting of the Technical Steering Committee (TSC) Board

Tuesday, June 19th 2018
11:00am ET

Meeting Logistics

- https://www.uberconference.com/jeff_ef

- United States : +1 (510) 224-9559 (No PIN needed).

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THE LINUX FOUNDATION

Agenda

- TSC call for nominations complete
  - Jeff will be sending out a google voting form to current TSC members this week. TSC members have till the end of the month to cast their votes
- ISC is next week, we will have a booth
- v1.3.5 release
  - went out last Wednesday (wahoo)
- gcc8 variant for next release?
  - gcc 8.1 released last month
  - question on github if we'll include that for 1.3.6…any objections?
- presentation on ohpc installation using ansible

OpenHPC Automation with Ansible

Baptiste Gerondeau, Takeharu Kato, Renato Golin

OpenHPC TSC meeting, 18th June 2018

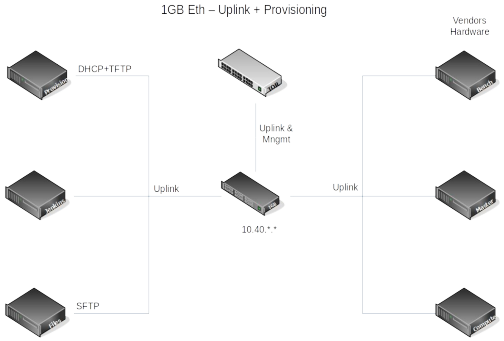


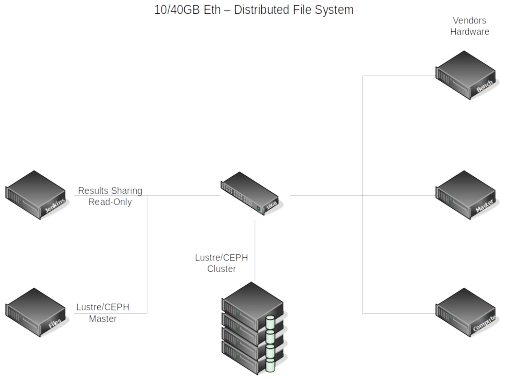


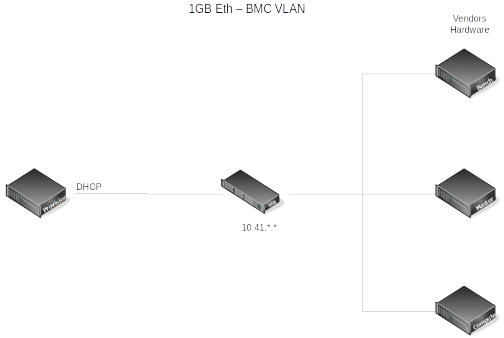
The HPC-SIG Lab

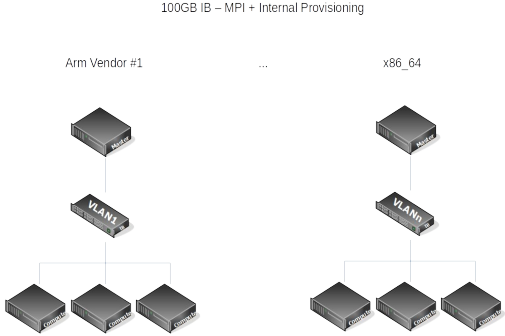
Network Layout

- Flat Ethernet
  - Uplink/Provision
- BMC subnet (VLAN)
- File System subnet
  - 10/40GB Ethernet
  - Lustre/Ceph (future)
- MPI subnet
  - 100GB InfiniBand
  - Slave provision



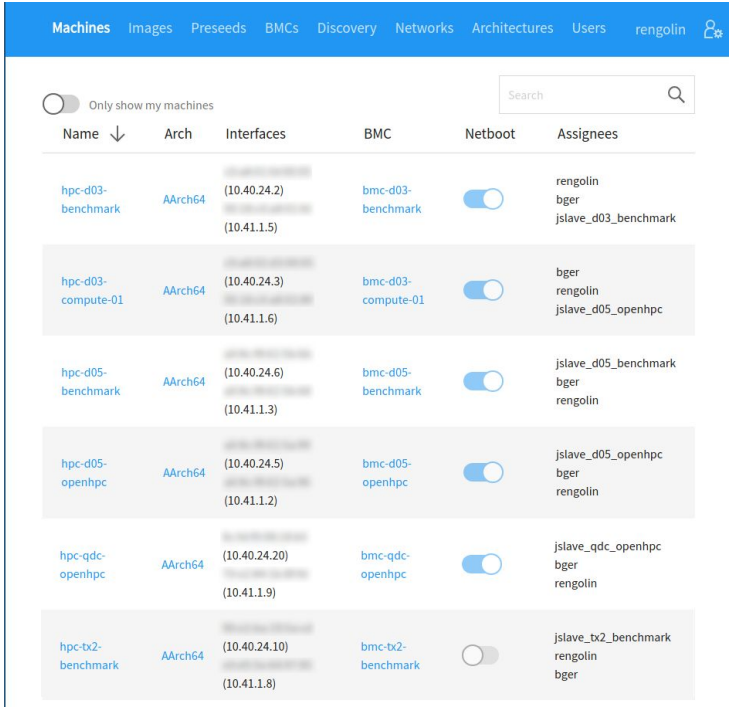






The HPC-SIG Lab

- Stability & Repeatability
  - Critical external components cached locally
  - Strict migration plans (staging)
- Close-to-production environment
  - Hardware and firmware updated frequently
- Upstream technology (reproducibility)
  - All components are open source / upstream
- Vendor isolation (hardware, results)
  - VPN, Provisioner, Jenkins, SSH control



Open Source and Upstream

Lab admin & tools:

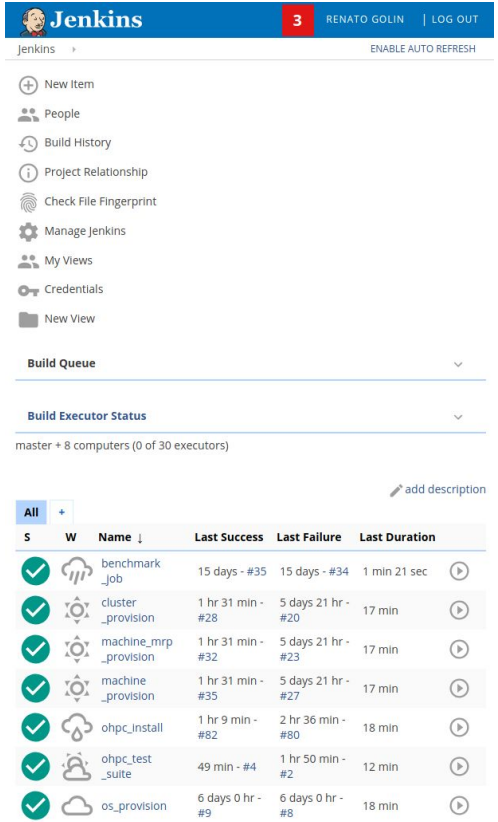
- Jenkins: https://jenkins.io/
- Mr-Provisioner: https://github.com/Linaro/mr-provisioner

Lab Automation:

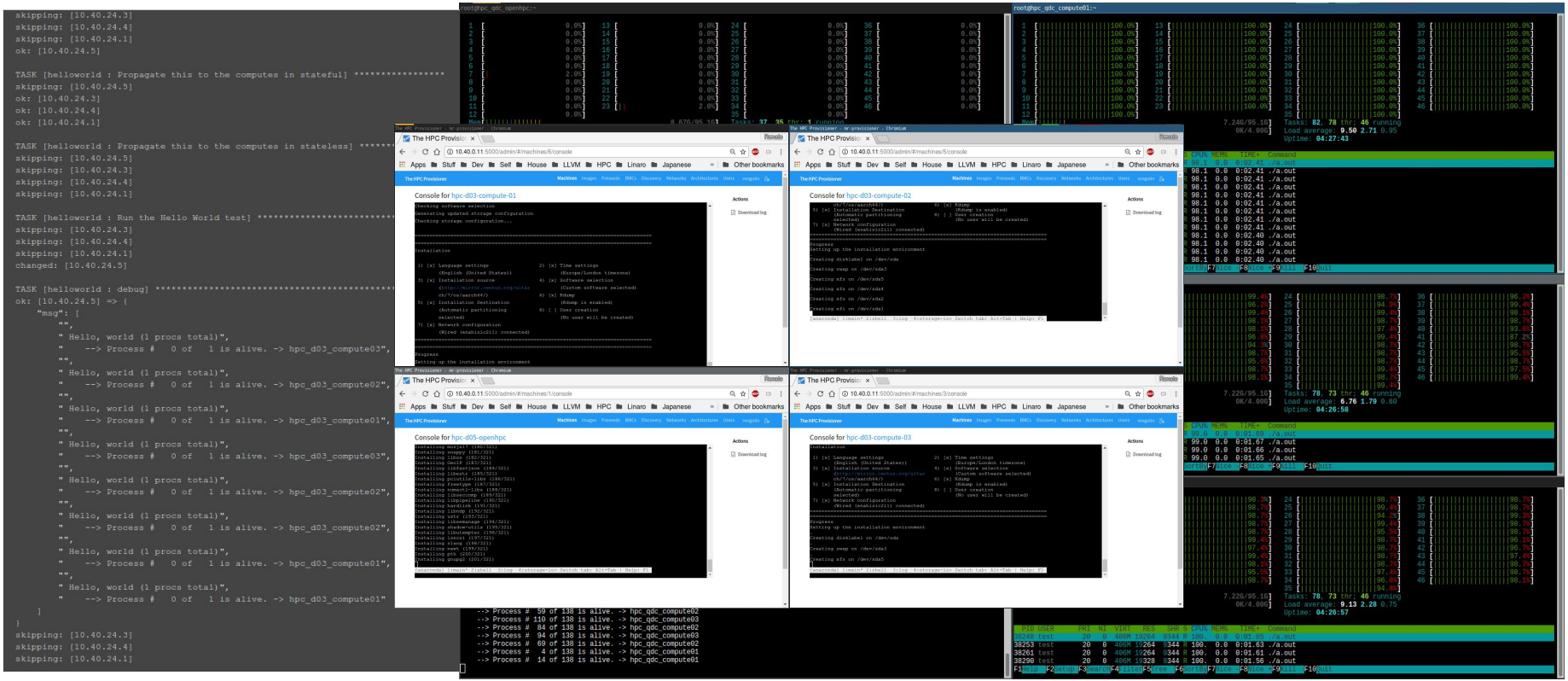
- https://github.com/Linaro/ans_setup_jenkins
- https://github.com/Linaro/mr-provisioner-role
- https://github.com/Linaro/mr-provisioner-kea-dhcp4-role
- https://github.com/Linaro/ansible-role-mr-provisioner
- https://github.com/Linaro/mr-provisioner-client

HPC-specific automation:

- https://github.com/Linaro/hpc_lab_jenkins
- https://github.com/Linaro/hpc_deploy_benchmarks
- https://github.com/Linaro/benchmark_harness
- https://github.com/Linaro/ansible-playbook-for-ohpc



Screenshots



Test Suite

Initial tests (green) to get the job going.

The plan from now on is:

1. Complete the list of packages working on Arm
2. Add an option for *additional* packages, so we can test them before sending upstream
3. Use that option for non-CI tests, like longer runs of the same packages, benchmarks, etc.

Libraries	
Adios	disabled
Boost	enabled
Boost MPI	enabled
FFTW	enabled
GSL	disabled
HDF5	disabled
HYPRE	disabled
MFEM	disabled
Metis	disabled
mpi4py	disabled
MUMPS	disabled
NetCDF	disabled
Numpy	disabled
OPENBLAS	enabled
PETSc	disabled
PHDF5	disabled
PLASMA	enabled
PNetCDF	disabled
PTScotch	enabled
R base package	disabled
ScaLAPACK	disabled
Scipy	disabled
Scotch	enabled
SLEPc	disabled
Superlu	disabled
Superlu_dist	disabled
Trilinos	disabled

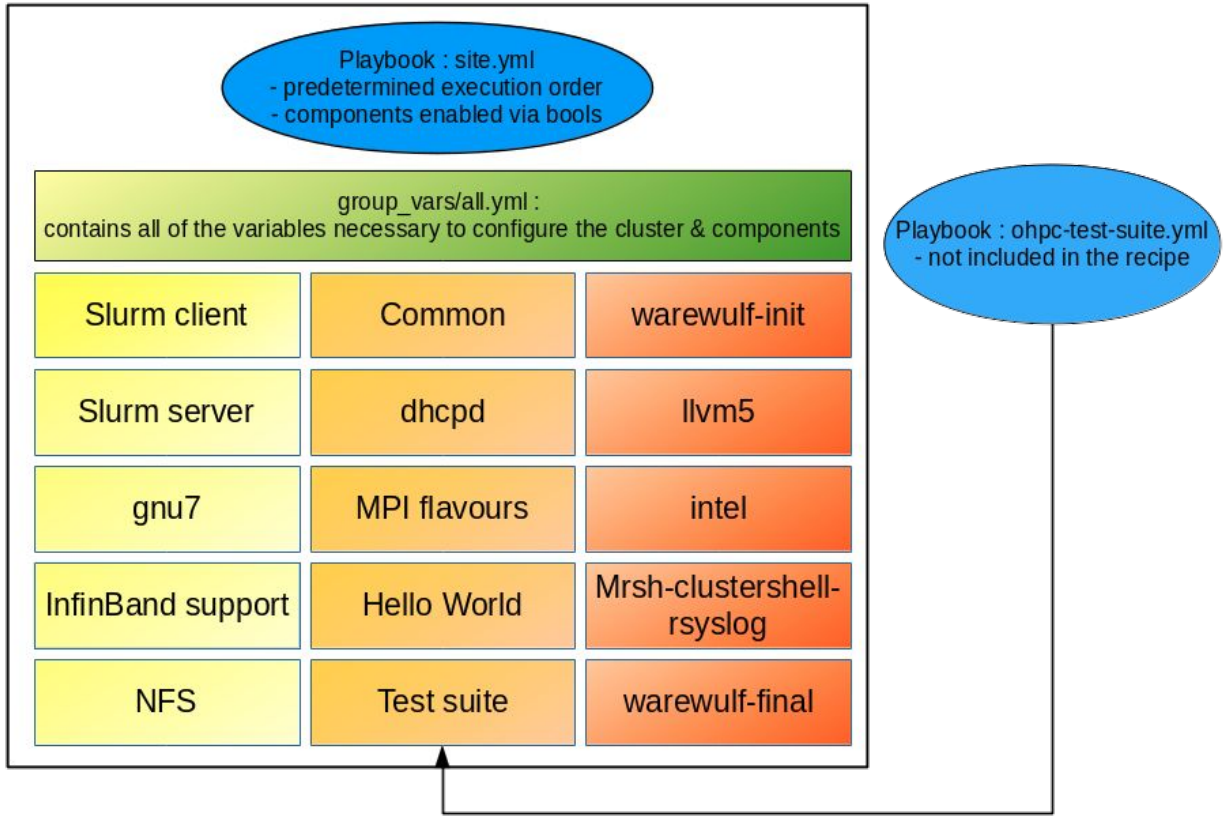
Dev Tools	
Autotools	enabled
CMake	enabled
CILK	disabled
EasyBuild	disabled
hwloc	enabled
TBB	disabled
Valgrind	enabled

Performance Tools	
IMB	disabled
Likwid	disabled
mpiP Profiler	disabled
Papi	disabled
Scalasca	disabled
TAU	disabled

Runtimes	
OCR	disabled

User Environment	
Compilers	enabled
Lustre client	disabled
Modules	enabled
MPI	enabled
Munge	enabled
OOM	enabled
Packaging tests	enabled
RMS test harness	enabled

Ansible Roles



Upstreaming our work

Stage #1:

- Generate from LaTex sources at the same time as recipe.sh
  - One bundle per recipe (warewulf/xcat, slurm/pbs, centos/suse, arm/x86)
  - Provide as a zip/tar file on the docs package, like the recipes
- Problems:
  - Still need to install package to get recipes that will install OpenHPC
  - Hard to collaborate upstream on the recipes themselves

Stage #2:

- Take the zip/tar file and push to a git repository
  - Use one branch per recipe (so people can checkout the right one)
  - Tag & release binaries (so people don't need to git clone)
- Issues and pull requests to the Ansible repo backported to OpenHPC LaTex sources

Thank You!



